Karthik Voruganti

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Education

Rutgers University – New Brunswick Expected May 2028
Bachelor of Science in Electrical & Computer Engineering | Minors: Mathematics, Computer Science
Relevant Coursework: Software Engineering, Computer Architecture, Machine Learning, Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Applied Algebra, Probability Theory, Programming Methodology

Experience

Mind360 AI | Software Engineer Sunnyvale, CA | May 2026 – Aug 2026

- Integrated 8 prediction-market venues using concurrent API requests, rate limiting, and live order-book validation to handle venue-specific limits and reject stale prices before execution.
- Extended a Python adversarial-testing engine with checkpoint recovery and SBOM/policy validation to resume interrupted runs, skip completed tests, and prevent incompatible results from merging.
- Built an on-screen text capture and simplification workflow with Electron, PaddleOCR, and Spring Boot; handled failures and edge cases across 3 model providers: Gemini, Groq, and OpenRouter.

Aviro AI (**Y Combinator S25**) | Machine Learning Research Intern San Francisco, CA | Jan 2026 – May 2026

- Co-developed 3 simulated enterprise environments and formulated 30 benchmark tasks to evaluate multi-step search, evidence retrieval, and cross-document reasoning.
- Benchmarked 8 frontier models and analyzed tool-call traces against source evidence to isolate retrieval failures, fabricated values, unsupported citations, and failures to abstain on unanswerable queries.
- Created 30 training tasks for reinforcement training of GPT-OSS-120b to engineer Ebla-1; contributed to a 7,200-rollout training run through training-job execution, reward evaluation, and checkpoint analysis.

Projects

Stratos

- Built a distributed edge-computing platform for drone fleet mapping and device coordination using FastAPI, Go, and MQTT/gRPC, with a registry of 46 board profiles across 3 hardware tiers.
- Implemented capacity-aware task dispatch over Redis Streams with agent-bound leases, stale-task reclamation, session validation, and idempotent result handling to recover work across intermittent connections.
- Integrated MAVLink telemetry from PX4/ArduPilot flight controllers; built retry-safe image uploads and restart-resilient NodeODM job orchestration to reconstruct multi-drone surveys into downloadable GeoTIFF maps.

Kairos

- Built a funding-discovery and application platform for student founders using FastAPI, Grants.gov search, and Amazon Bedrock agents to identify relevant programs and draft source-grounded applications.
- Achieved 85.7% retrieval recall on a 20-program benchmark using curated listings and rule-based eligibility filters; separated assessment, drafting, and independent review with Strands Agents.
- Implemented a fail-closed validation gate checking source references, protected fields, and numeric/entity allowlists to block unsupported drafts and explicitly flag missing information.

Fiducia

- Built a creator sponsorship platform with React, Fastify, and Prisma, connecting USDC escrow to 4 base-payment milestones: concept approval, final-cut approval, verified publication, and retention.
- Implemented OAuth and metric adapters for 4 social platforms; required 2 consecutive compliant YouTube checks before performance settlement and bound approvals to immutable deliverable hashes.
- Developed Solidity escrow contracts with Base deployment support to reserve campaign funds, anchor agreement hashes, enforce payout caps, reject duplicate releases, and refund unreleased balances.

Future360

- Built a prediction-market terminal spanning 8 venues, combining Python and REST/WebSocket feeds to compare contracts, detect arbitrage, and coordinate multi-leg order execution.
- Filtered false opportunities with rule-based and fuzzy matching plus resolution, liquidity, and fee checks; combined idempotency keys, persistent per-leg state, and runner leases with live quote revalidation and partial-fill reconciliation.

Skills

Languages: Python, Go, Java, C, C++, TypeScript, JavaScript, SQL, Solidity
Frameworks & Libraries: FastAPI, Spring Boot, Fastify, React, Next.js, Electron, Prisma, TensorFlow/Keras, scikit-learn, Pandas
Data & Protocols: PostgreSQL, MySQL, Redis Streams, REST, WebSockets, MQTT, gRPC, Protocol Buffers
Cloud: AWS (ECS/Fargate, RDS, SQS, S3, EventBridge, CloudWatch, Bedrock)
Tools: Docker, Kubernetes, Linux, Git, GitHub Actions